Assertions in Python

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What is an assertion?

An assertion is a **sanity check** that you can turn on or off when testing your program. Assertions in Python are used **to perform checks during program execution**, mainly for **debugging** purposes.

They are used to verify that a **certain condition is true** at a specific point in the code. If the **condition is not true**, Python raises an **AssertionError** exception.

Sintax:

```
assert condition, "Optional error message"
```

Example:

```
x = 5

assert x > 0, "x must be positive"
```

Purpose of assertions:

✅ Ensure assumptions are true during development

✅ Help detect bugs early

❌ Not for handling user input or expected runtime errors

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The optimize (-O) parameter

If you wrote an important check using assert, that **check will never be executed** when running in **-O mode**.

Example:

```
assert password != "", "Password cannot be empty"
```

Executing:

`python -O script.py` -> assertions ignored

`python script.py` -> assertions not ignored

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